



A novel transfer gear life prediction method by the cross-condition health indicator and nested hierarchical binary-valued network

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ABSTRACT

As one of the crucial techniques in prognostics and health management, the long-term remaining useful life (RUL) prediction of gears is essential for maintenance decisions in engineering applications. However, most previous RUL prediction approaches may have unacceptable prediction performance due to insufficient prior life-cycle data and variable operating conditions. In this paper, a novel transfer life prediction method is proposed for gear RUL estimation under different working conditions. First, a health indicator (HI) transfer construction framework, QFMDCAET, is developed for generating cross-condition HIs under various working conditions based on a quadratic function-based multi-scale deep convolutional auto-encoder network and multi-kernel maximum mean discrepancy. With the acquired gear HIs, the nested hierarchical binary-valued network (NHBN) is designed to estimate the gear RUL. In NHBN, a Gumbel function is applied to improve the memory ability of NHBN. Moreover, a nested hierarchical mechanism is also used to fully utilize the sequence information of the cell state in NHBN. The experimental results illustrate the effectiveness of the proposed NHBN in gear RUL prediction, especially for long-term RUL prediction. Lastly, the comparison results indicate that the proposed NHBN-based RUL prediction method is superior to other conventional and advanced RUL prediction methods.

1. Introduction

Gear, as a significant transmission component, has been widely applied in the industrial community. Gear plays the role of transmitting torque and motion in some equipment, such as mining equipment, wind turbine gearbox, and aero-engine driving system [1,2]. Due to the hard-working environment of the above equipment, the gear is prone to failure after long-term operation. Pitting, scuffing, and the broken tooth will occur in a closed-gear transmission system, while abrasive wear or other faults will occur in an open-gear transmission system. These gear faults can lead to equipment downtime or even serious casualties [3–6]. Therefore, it is necessary to predict the remaining useful life (RUL) of

gear to improve the safety and reliability of equipment, which can reduce downtime and avoid frightful casualties.

In the prognostics and health management (PHM) field, RUL prediction of rotating machinery has been widely investigated in the academic and industrial fields. The RUL prediction can help engineers better understand the health status of equipment for ensuring its safe operation. With the estimated RUL, engineers can develop an appropriate equipment operation strategy to extend its service life and improve its reliability.

The current RUL prediction methods can be roughly divided into two categories: physics-based method and data-driven method [7–9]. The former can obtain accurate RUL prediction results due to known fault

Abbreviations: RUL, remaining useful life; HI, health indicator; NHBN, nested hierarchical binary-valued network; PHM, prognostics and health management; QFMDCAE, quadratic function-based deep convolutional auto-encoder; QFMDCAE, quadratic function-based multi-scale deep convolutional auto-encoder network; MK-MMD, multi-kernel maximum mean discrepancy; LSTM, long short-term memory; ON-LSTM, ordered neurons long short-term memory; HBLSTM, hierarchical binary-valued LSTM; RNN, recurrent neural network; RKHS, reproducing kernel Hilbert space; RTRL, real-time recurrent learning; FPT, first prediction time; Mon, monotonicity; Cor, correlation; Rob, robustness; CI, composite index; PDFs, probability density functions; GRU, gated recurrent unit; MMALSTM, macro-micro attention mechanism LSTM; GAU, gated attention unit; GDAU, gated dual attention unit; MAE, mean absolute error; NRMSE, normalized root mean square error; MAPE, mean absolute percentage error; RMSE, root mean square error; NHLSTM, nested hierarchical LSTM.

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mechanisms and physical information [10,11]. However, some poor prediction results may be obtained when the tested object is a complex mechanical system. The data-driven method has been explored in recent years to compensate for the disadvantages of the physics-based method [12,13]. This method has the ability to directly mine significant degradation feature information from the acquired monitoring data of complicated systems. Thus, the data-driven approach requires less prior knowledge and degradation mechanism than the physics-based method. Moreover, with the rapid development of sensing, computer, and big data technologies, artificial intelligence-based RUL prediction methods have become the mainstream method in data-driven methods.

In artificial intelligence-based RUL estimation framework, there are three main parts: data acquisition, degradation feature extraction (health indicator construction), and life prediction (regression prediction or time-series prediction). In the first part, the full life-cycle datasets that contain degradation information of machine are acquired by data collection system. In the second part, constructing a suitable health indicator (HI) is extremely significant for time-series-based RUL prediction method. The constructed HI can describe the health state of machine and improve the prediction accuracy of RUL prediction methods. The supervised and unsupervised HI construction approaches for rotating machinery have been studied in the past several years. For supervised HI construction method, the HI construction model is trained via the designed HI labels. Guo et al. [14] employed convolutional neural network to construct HI of bearing. The outlier regions in the obtained HI were removed by an outlier region correction technique. Qin et al. [15] presented a HI construction method for generating HI of rotating machinery based on a novel supervised multi-head self-attention auto-encoder. Chen et al. [16] developed a quadratic function-based deep convolutional auto-encoder (QFDCAE) network to generate bearing HIs from raw vibration signals. For unsupervised HI construction method, the HI can be generated without any HI labels. Guo et al. [17] proposed a multi-scale convolutional auto-encoder network to extract the HI of a machine via an unsupervised feature learning manner. Li et al. [18] developed a shape-constrained neural data fusion network to construct HI from sensor data with a novel loss function. He and Guo [19] built an improved Gaussian mixture model named Hellinger distance-based regularized Gaussian mixture model to construct bearing HIs from run-to-failure datasets.

According to the abovementioned works about HI construction, these methods need abundant life-cycle datasets for model training, and the trained model cannot be directly applied for HI construction under other working condition. This is because that the data under different working conditions have various data distribution. Hence, to achieve HI transfer construction, a quadratic function-based multi-scale deep convolutional auto-encoder network (QFMDCAE) and multi-kernel maximum mean discrepancy (MK-MMD) are applied to design a gear's HI transfer construction framework named QFMDCAET and generate gear's HIs under multiple working conditions.

In the life prediction part, the pattern recognition-based method can learn the potential relationship between the measured datasets and the designed RUL labels via an end-to-end prediction framework. To better mine significant degradation information by integrating attention mechanism into the RUL prediction framework, Zhang et al. [20] developed a novel integrated multi-head dual sparse self-attention network to predict the RULs of turbofan engines. Xu et al. [21] proposed a new RUL prediction approach based on an improved transformer model for estimating the RULs of aero-engines. In order to effectively leverage the superiority of gating mechanism, Mo et al. [22] presented a transformer encoder-based model for RUL estimation of the machine. Wang et al. [23] designed a novel gated graph convolutional network to extract degradation features from multi-sensor data for estimating machine's RUL.

The pattern recognition-based approach can achieve acceptable prediction accuracy when abundant labeled training datasets exist. However, in practice, obtaining sufficient full life-cycle datasets could

be difficult, expensive, and time-consuming, especially when the life-cycle datasets are incomplete. The prediction performance of the pattern recognition-based RUL prediction approaches will drastically decrease when insufficient life-cycle datasets are used for training these neural network-based methods. In addition, these approaches cannot obtain the degradation rate of the testing object. This is mainly because they are end-to-end frameworks that can only predict the RUL of the testing object from the monitoring datasets.

The time-series prediction method is proposed to address the abovementioned problems and estimate the RUL of a machine from insufficient life-cycle datasets. It can learn the degradation trend from the known HIs and predict the unknown HIs step by step. The predicted HIs could describe the degradation rate of the machine, while the RUL can be calculated when the predicted HI exceeds the preset failure threshold. In recent years, some time-series prediction approaches have been explored for the RUL prediction of a machine. Considering small sample prediction, Pan et al. [24] built a two-stage approach based on feedback extreme learning machine model for short-term RUL prediction of rolling bearing. Towards the prediction error superposition problem, Lu et al. [25] proposed a deep adversarial learning prediction framework based on long short-term memory (LSTM) and generative adversarial network for life prediction of bearing. To improve the feature learning capacity of LSTM in time-series estimation approach via attention mechanism, Xiang et al. [26] developed an LSTM network with weight amplification to estimate gear RUL based on a fused HI. In order to enhance the memory capacity of recurrent neural network for RUL prediction, Yan et al. [27] proposed a gear RUL prediction approach based on an ordered neurons long short-term memory (ON-LSTM) network. Chen et al. [28] proposed a gated adaptive hierarchical attention unit neural network to estimate the RULs of servo motors. Zhou et al. [29] designed a consolidated memory gated recurrent unit network to predict the RULs of bearings. According to the literature review, the RUL estimation method can evaluate the health state of the mechanical object with the predicted HIs. Moreover, it can also predict the machine's RUL with limited life-cycle samples or incomplete life-cycle samples.

Although the pattern recognition-based method has achieved superior predictive performance, the prediction accuracy may significantly decrease when the testing data are acquired from other working conditions. This is because the testing data distribution is different from the training data one. Consequently, the trained life estimation method cannot be directly applied to the RUL prediction of the testing data. Transfer learning technology has been utilized for RUL prediction to improve the prediction performance of the RUL prediction method under various working conditions.

In the past few years, some transfer RUL prediction methods based on pattern recognition have been investigated for RUL estimation under variable working conditions. Through the generative adversarial mechanism, Ragab et al. [30] presented a contrastive adversarial domain adaptation model to predict the cross-domain RUL of a machine. Zhuang et al. [31] developed a metric adversarial domain adaptation method for bearing RUL estimation under different working conditions. Hu et al. [32] built a Wasserstein distance-based weighted domain adversarial model to obtain the bearing's transfer life. Using maximum mean discrepancy, He et al. [33] designed a transferable neural network for cross-domain RUL prediction. To extract fine-grained features in various degradation stages, Zhang et al. [34] developed a variational local weighted deep sub-domain adaptation network for RUL prediction under different working conditions.

From the abovementioned literature, the transfer RUL prediction approaches based on pattern recognition can estimate the RULs of target objects in the target domain. However, these trained models cannot simultaneously evaluate the health status of the target object and estimate the RUL of the target object in the target domain with limited run-to-failure datasets or even incomplete life-cycle datasets. Thus, it is necessary to explore a transfer RUL prediction framework based on time-series prediction for RUL estimation under various working conditions.

To address this issue, an HI transfer construction framework should first be developed for extracting HIs under different working conditions. Next, a time-series-based RUL prediction network can be carefully designed to improve prediction accuracy with a reliable confidence interval. After obtaining the gear cross-condition HIs, the designed time-series prediction method can be trained to learn the degradation trend from the known gear HIs. Then, unknown gear HIs that can describe the health status of gear can be predicted via the trained method. Finally, the RUL of the gear is calculated when the predicted HI reaches the preset failure threshold.

To ensure the prediction accuracy of the time-series prediction method, the RUL prediction method should have the capacity for long-term RUL prediction. Nowadays, the LSTM model and its variants are broadly utilized for RUL prediction, especially for long-term RUL prediction. Such models can filter some unwanted information from current input data and historical retained data via the specific gate mechanism. To update the weight and bias parameters of these networks by stochastic gradient descent algorithm and mimic the gate in the LSTM network, the sigmoid activation function is applied, which can output soft values between 0 and 1. However, by exploring the data distribution of the output gating values in LSTM from many experiments and some related literature [35,36], it can be concluded that the gating values around 0.5 have a higher probability than the gating values around 0 or 1. This phenomenon disobeys the logic of the gating mechanism, and the RUL prediction performance of the LSTM-based method may be affected if most gating values are ambiguous. Apart from the gate mechanism, the sequence information of neurons in LSTM is also not considered, meaning that the neurons in LSTM are disordered, and the time-varying information cannot be effectively learned.

To address the above problems, obtain gating values around 0 or 1 with high probability, and fully utilize the sequence information of neurons, an improved LSTM-based network should be urgently developed. Therefore, a novel variant of LSTM called nested hierarchical binary-valued network (NHBN) is presented for gear's RUL prediction with the generated cross-condition HIs. In this network, a Gumbel function is applied to approximate the sampled values of the Bernoulli distribution. Therefore, the output values of the input and forget gates in the proposed NHBN can be closer to 0 or 1. In other words, the modified gates in the proposed network can retain more long-term historical information and update additional short-term current information. Based on the output values of these two gates, a hierarchical mechanism is designed to effectively leverage the sequence information of neurons. Moreover, a hierarchical binary-valued LSTM (HBLSTM) is developed to update the cell state via three different hierarchies: high cell hierarchy, middle cell hierarchy, and low cell hierarchy. However, the simple addition operation in the updating formula of the middle cell hierarchy may lose some significant sequence information in the training process.

To deal with the aforementioned problem and fully utilize the sequence information of neurons, a nested hierarchical mechanism that uses the hidden state of an inner HBLSTM to update the cell state in the middle cell hierarchy of the outer HBLSTM is designed. Finally, the developed NHBN is constructed based on the Gumbel function and the nested hierarchical mechanism. With the advantages of the Gumbel function and the nested hierarchical mechanism, the developed NHBN can learn the time-varying trend information from the known HI points and accurately predict RUL with the trained model. Moreover, a new gear RUL prediction methodology under different working conditions is proposed based on the QFMDCAET and NHBN. The superiority of the proposed RUL prediction method is proven via comparison with other classic and advanced RUL prediction approaches using the gear monitoring datasets under different working conditions. The main contributions of this work are as follows:

- (1) To restrict the gate output values closer to the boundary regions and improve the memory capacity of LSTM, a Gumbel function, which can approximate the sampled values of the Bernoulli

distribution, is applied. This function can generate additional gating values around 0 or 1 compared to the conventional sigmoid function in LSTM.

- (2) The sequence information of neurons is ignored in the traditional LSTM. In order to effectively leverage the sequence information, an HBLSTM is first constructed to divide the sequence information of the cell state into three cell hierarchies. To further fully utilize the sequence information in the middle cell hierarchy, a nested hierarchical mechanism is designed to update the cell state in the middle cell hierarchy with another HBLSTM.
- (3) Using the Gumbel function and the nested hierarchical mechanism, the NHBN is built based on HBLSTM, and the related hierarchical updating strategy is derived. The predictive results demonstrate that the proposed network has better prediction performance than LSTM and its variants.
- (4) To realize the time-series-based RUL prediction across various working conditions, a gear HI transfer construction framework named QFMDCAET is first constructed to obtain gear cross-condition HIs. Then, a time-series prediction approach is developed for RUL prediction with the transfer HIs and NHBN, which has a stronger prediction capacity than other classic and latest time-series prediction approaches.

2. Theoretical foundation

2.1. Quadratic function-based deep convolutional auto-encoder

The QFDCAE network was first proposed to generate bearing HIs [16]. The structure of the QFDCAE is shown in Fig. 1. There are several convolutional and pooling layers in the encoder and decoder of the QFDCAE network. These layers can mine potential degradation information from the high-dimensional input data. Therefore, this network has the capacity to automatically extract HI from the raw monitoring data with the designed HI label.

Form Fig. 1, it can see that the fully connected layers in AE have been replaced by the convolutional and pooling layers in QFDCAE. Moreover, the output of the encoder in AE is also not constrained by the regularization term in loss function, which will lead to the extracted HIs have poor degradation trend. However, for QFDCAE, a special quadratic function is designed to generate HI labels. In the training process of QFDCAE network, these HI labels can guarantee that the constructed HIs have favorable degradation trends by adding a constraint term into the loss function.

2.2. Long short-term memory network

As a special type of recurrent neural network (RNN), LSTM was proposed by Hochreiter and Schmidhuber [37]. LSTM is explicitly designed to address the long-term dependency problem by the gate mechanism. The detailed call structure of the LSTM is demonstrated in Fig. 2, and the special derivation formulas are defined as follows:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f) \quad (1)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i) \quad (2)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o) \quad (3)$$

$$\mathbf{g}_t = \varphi(\mathbf{W}_g \mathbf{x}_t + \mathbf{U}_g \mathbf{h}_{t-1} + \mathbf{b}_g) \quad (4)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t \quad (5)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \varphi(\mathbf{c}_t) \quad (6)$$

where $\sigma(\cdot)$ and $\varphi(\cdot)$ are nonlinear activation functions; $\mathbf{x}_t$ is the input data at the current time; $\mathbf{h}_{t-1}$ is the hidden state at the previous time; $\mathbf{f}_t$, $\mathbf{i}_t$, and $\mathbf{o}_t$ are the output gating values of the forget gate, input gate, and

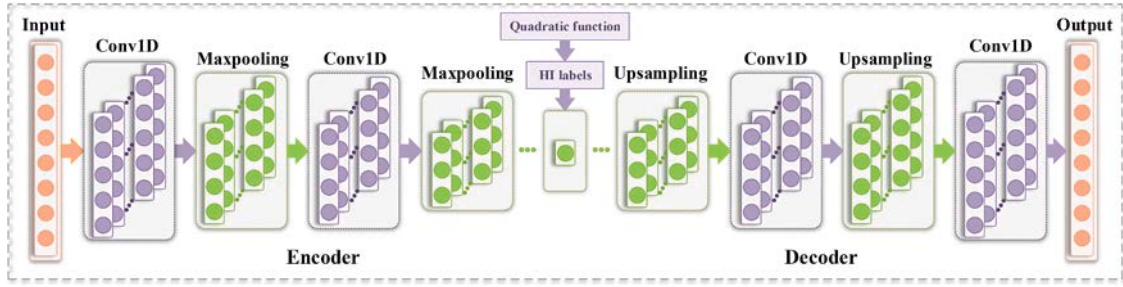


Fig. 1. The structure of the QFDCAE network.

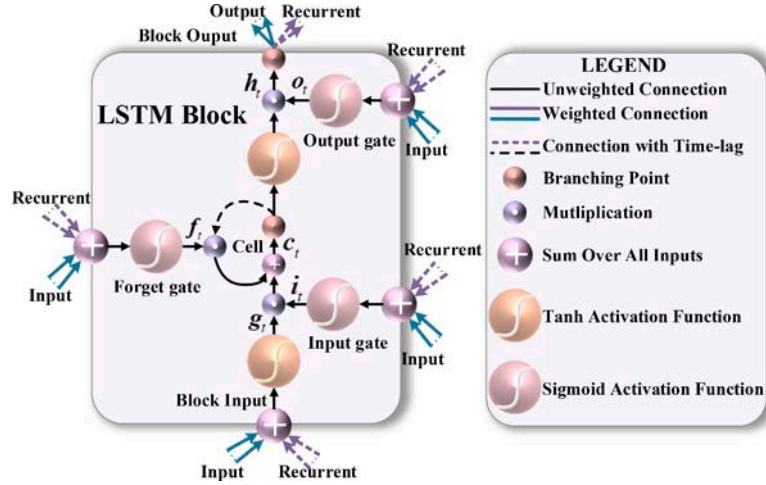


Fig. 2. The cell structure of the LSTM network.

output gate, respectively; g_t is the candidate state at the current time; W , U , and b are the weight and bias parameters that can be learned via gradient descent algorithm; c_t is the cell state at the current time; h_t is the hidden state at the current time, and $\odot$ is the element-wise product.

3. The proposed transfer RUL prediction methodology

3.1. Gear health indicator transfer construction framework

The QFDCAE network introduced in the previous section can construct HIs with favorable monotonicity from raw vibration signals. However, since only a single convolutional kernel is used to extract

degradation features in the convolutional layer, the degradation feature extraction capacity of QFDCAE is restricted. In order to deal with this restriction, a multi-scale feature extraction module should be applied to extract various abstract features from input data in the QFDCAE network. Fortunately, the inception module of GoogleNet provides a viable way to extract multi-scale features via several convolutional kernels with different scales. The QFMDCAE is constructed for HI construction of gear by replacing the convolutional layer in QFDCAE with the inception module. The structure of this network is shown in Fig. 3. The detailed network architecture of QFMDCAE is listed in Table 1. From this table, it can know that there are 26 layers in the constructed QFMDCAE network, including one input layer, nine inception modules,

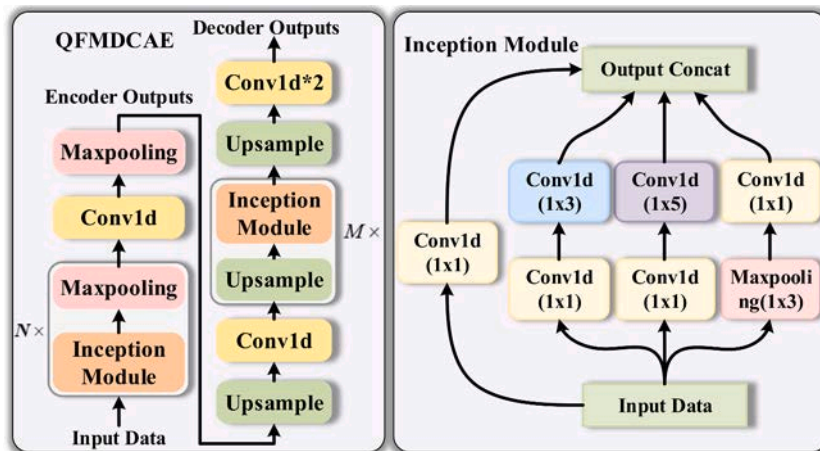


Fig. 3. The structure of the QFMDCAE network.

Table 1

The detailed network architecture of the QFMDCAE network.

Module	Layer	Block	Operator	Filter size (Scale factor)	Input size	Output size
Decoder	26	C4	1-D Convolution	3	(1,5000)	(1,5000)
	25	C3	1-D Convolution	3	(64,5000)	(1,5000)
	24	U6	Up-Sampling	5	(64,1000)	(64,5000)
	23	IM9	Inception Module	1/1,3/1,5/3,1	(128,1000)	(64,1000)
	22	U5	Up-Sampling	5	(128,200)	(128,1000)
	21	IM8	Inception Module	1/1,3/1,5/3,1	(256,200)	(128,200)
	20	U4	Up-Sampling	5	(256,40)	(256,200)
	19	IM7	Inception Module	1/1,3/1,5/3,1	(128,40)	(256,40)
	18	U3	Up-Sampling	5	(128,8)	(128,40)
	17	IM6	Inception Module	1/1,3/1,5/3,1	(64,8)	(128,8)
	16	U2	Up-Sampling	4	(64,2)	(64,8)
	15	C2	1-D Convolution	1	(1,2)	(64,2)
	14	U1	Up-Sampling	2	(1,1)	(1,2)
Encoder	13	P6	Max-Pooling	2	(1,2)	(1,1)
	12	C1	1-D Convolution	1	(64,2)	(1,2)
	11	P5	Max-Pooling	3	(64,8)	(64,2)
	10	IM5	Inception Module	1/1,3/1,5/3,1	(128,8)	(64,8)
	9	P4	Max-Pooling	3	(128,40)	(128,8)
	8	IM4	Inception Module	1/1,3/1,5/3,1	(256,40)	(128,40)
	7	P3	Max-Pooling	3	(256,200)	(256,40)
	6	IM3	Inception Module	1/1,3/1,5/3,1	(128,200)	(256,200)
	5	P2	Max-Pooling	3	(128,1000)	(128,200)
	4	IM2	Inception Module	1/1,3/1,5/3,1	(64,1000)	(128,1000)
	3	P1	Max-Pooling	3	(64,5000)	(64,1000)
	2	IM1	Inception Module	1/1,3/1,5/3,1	(1,5000)	(64,5000)
	1	Input	1-D Signal	5000	/	5000

four convolutional layers, six max-pooling layers, and six up-sampling layers. The encoder is composed of the first 13 layers, while the decoder consists of the last 13 layers.

The current HI construction methods may need sufficient run-to-failure data to train the backbone networks in these methods. However, it is difficult to obtain many run-to-failure data in real industry. Moreover, the trained HI construction network under a single working condition cannot be directly utilized to construct HI under another working condition. This is mainly because the data distributions of these two datasets are different. The constructed HI may have poor monotonicity, and the failure threshold for RUL prediction cannot be uniformly set when the trained model is directly used to construct HI under different working conditions. Thus, the knowledge transfer technology is applied to develop a gear HI transfer construction framework called QFMDCAET and construct gear HI with superior monotonicity and uniform failure threshold.

The structure of this framework is demonstrated in Fig. 4. The QFMDCAE is employed to extract degradation features from the input data of source and target domains. After extracting the source-domain and target-domain features, the MK-MMD is used to measure the distribution discrepancy of extracted features. The extracted two domain features are simultaneously mapped into the reproducing kernel Hilbert space (RKHS) by linear combination of multiple kernels, and then the difference between the two domain features is measured by the mean distance in the mapping space. Then, the obtained mean distance values will be reduced by adding a regularization term in loss function. This means that the distribution discrepancy between the source domain features and target domain features will be minimized via MK-MMD, and the domain invariant features can be extracted by the trained network. Finally, the trained QFMDCAE network is utilized to construct the gear's HI of the target domain from the life-cycle data in the target domain. The acquired gear HI has good monotonicity and can express the degradation trend of the gear. According to the obtained gear HIs under multiple operating conditions, the failure threshold significant for the time-series prediction method can also be uniformly preset for RUL prediction in practical application.

3.2. Nested hierarchical binary-valued network

The LSTM core is the cell state that can remember some significant information from current input data and previously retained data via the input and forget gates. However, most of the output gating values of the input and forget gates are around 0.5, which is not aligned with the physical gate mechanism (e.g., closing or opening). This may result in some unwanted information being retained or some significant information being lost in the cell state. In addition, the sequence information of neurons is also not considered in the traditional LSTM. These drawbacks will weaken the long-term memory ability of LSTM.

To improve the long-term memory capacity of the conventional LSTM, a novel variant of LSTM named HBLSTM was first designed. In the HBLSTM network, the sigmoid function of the input and forget gates are replaced by the Gumbel function, which can obtain additional gating output values around the boundary regions. This would conform to the original intent of the gating mechanism in LSTM. The HBLSTM can divide the cell state into three cell hierarchies (e.g., high cell hierarchy, middle cell hierarchy, and low cell hierarchy). However, the ordinary addition operation of the updating formula in the middle cell hierarchy may lose some long-term memory or short-term memory information in the updating process of the cell state. Hence, another inner HBLSTM is regarded as a learned function to mine more long-term memory or short-term memory information from the middle cell hierarchy of the outer HBLSTM and address this drawback while fully utilizing the sequence information of the cell state in the middle cell hierarchy. Finally, the NHBN is developed via the outer HBLSTM and the inner HBLSTM. The cell structures of HBLSTM and NHBN are demonstrated in Figs. 5 and 6, respectively.

In the original LSTM, the sigmoid function is employed to generate some soft values between 0 and 1 for mimicking the gate. However, by analyzing the data distribution of the gating values of the input and forget gates, most gating values are closer to the middle point of the sigmoid function values. This is not aligned with the design logic of the gate mechanism. To make the gates in LSTM toward binary-valued gates, a Gumbel function is employed to generate additional gating values around boundary regions. The Gumbel function is defined as follows:

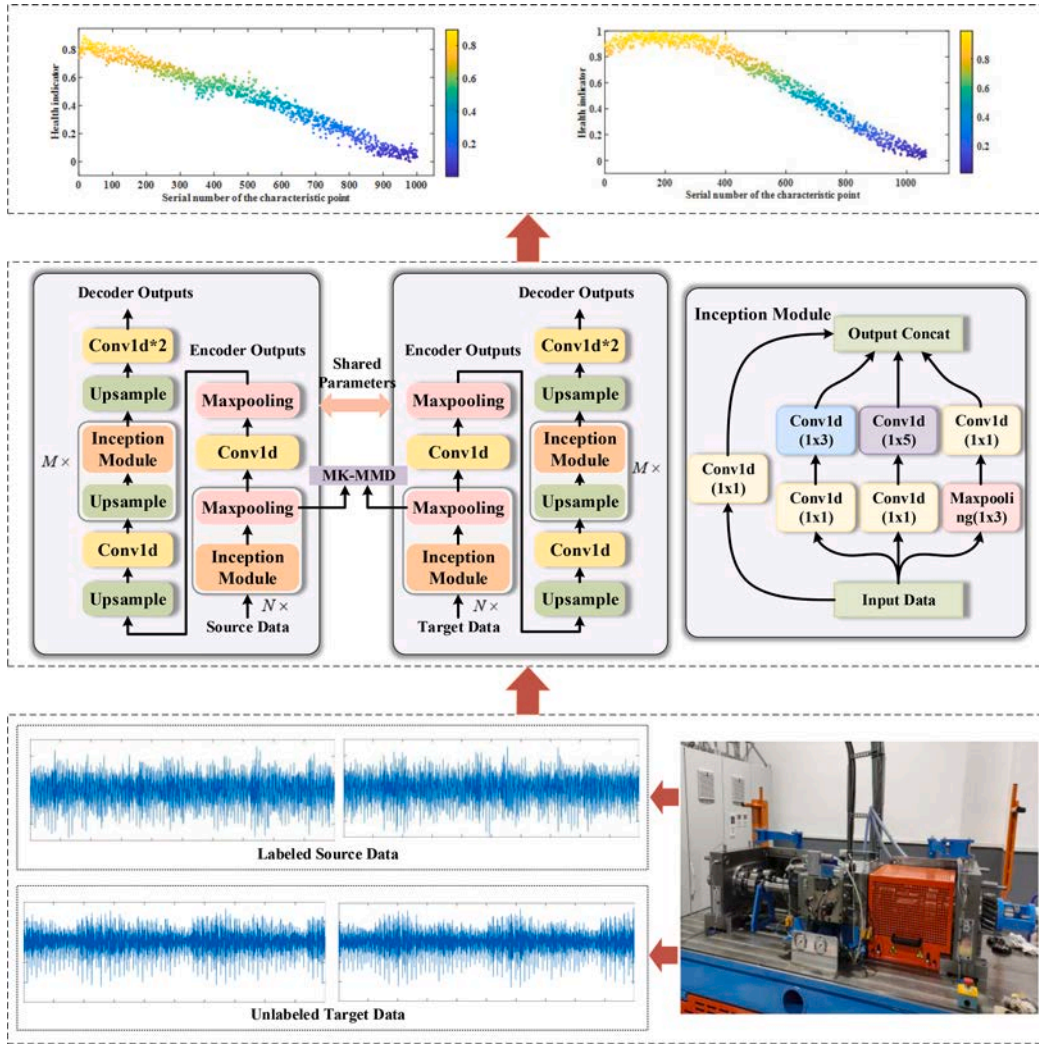


Fig. 4. The flowchart of gear HI transfer construction framework.

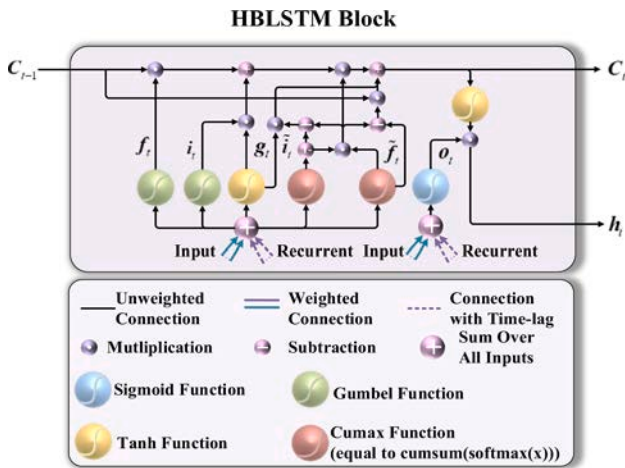


Fig. 5. The cell structure of HBLSTM.

$$R(x, \lambda) = \sigma\left(\frac{x + \log U - \log(1 - U)}{\lambda}\right) \quad (7)$$

where x is the input data, λ is a preset parameter ($\lambda > 0$), U is a vector where each value in U is independently sampled from Uniform (0, 1), and σ is the sigmoid function. The Gumbel function is efficient in approximating the sampled values of discrete distribution (e.g., Bernoulli distribution) [36]. The Gumbel function is utilized to obtain the output values of the input and forget gates in this work, and these output values will update more significant information into the cell state of the HBLSTM. The Gumbel function can improve the long-term memory capacity of the designed HBLSTM. Forget and input gates of the HBLSTM are redefined as follows:

$$\mathbf{f}_t = R(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f, \lambda) \quad (8)$$

$$\mathbf{i}_t = R(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i, \lambda) \quad (9)$$

To effectively leverage the sequence information of neurons for further enhancing the long-term memory ability of HBLSTM, a hierarchical mechanism inspired by ON-LSTM [38] is designed. The master forget gate $\tilde{\mathbf{f}}_t$ and the master input gate $\tilde{\mathbf{i}}_t$ are calculated via the following formulas:

$$\tilde{\mathbf{f}}_t = \text{cumsum}(\text{softmax}(\mathbf{W}_{\tilde{f}} \mathbf{x}_t + \mathbf{U}_{\tilde{f}} \mathbf{h}_{t-1} + \mathbf{b}_{\tilde{f}})) \quad (10)$$

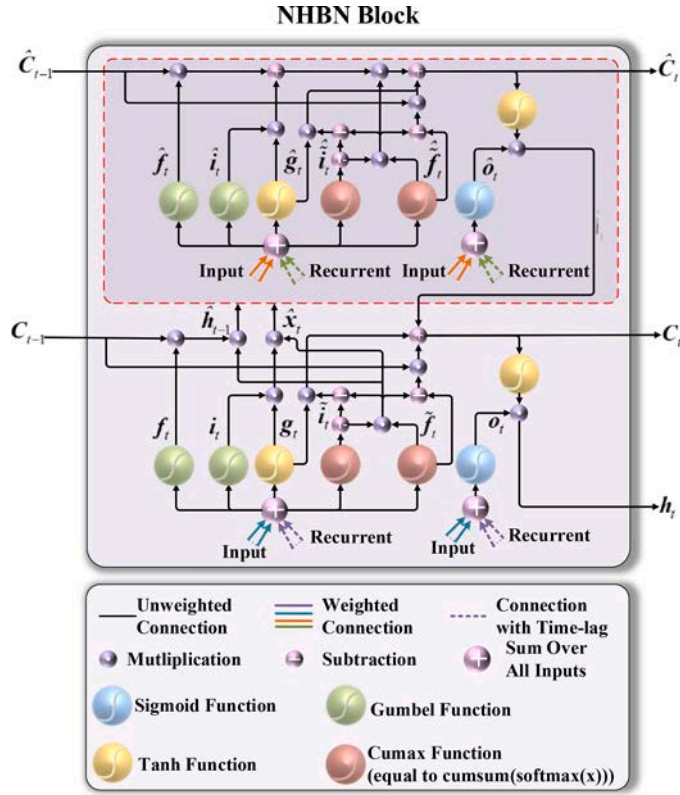


Fig. 6. The cell structure of NHBN.

$$\tilde{\mathbf{i}}_t = 1 - \text{cumsum}(\text{softmax}(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i)) \quad (11)$$

Three different hierarchies can be obtained by the following equations based on the outputs of the master forget gate and master input gate:

$$\omega_{t1} = \tilde{\mathbf{f}}_t \odot \tilde{\mathbf{i}}_t \quad (12)$$

$$\omega_{t2} = \tilde{\mathbf{f}}_t - \omega_{t1} \quad (13)$$

$$\omega_{t3} = \tilde{\mathbf{i}}_t - \omega_{t1} \quad (14)$$

where ω_{t1} is the middle cell hierarchy, ω_{t2} denotes the high cell hierarchy, and ω_{t3} is the low cell hierarchy. After that, the cell state $\mathbf{c}_t$ of HBLSTM can be updated using Eqs. (4), (8), and (9) as follows:

$$\mathbf{c}_t = \omega_{t2} \odot \mathbf{c}_{t-1} + \omega_{t1} \odot (\mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t) + \omega_{t3} \odot \mathbf{g}_t \quad (15)$$

However, the ordinary addition operation in Eq. (15) for the middle cell hierarchy may lead to some wanted memory information being lost in the updating process of the cell state of HBLSTM. To avoid losing memory information, a learned updating function is used to replace the addition operation for mining additional long-term memory or short-term memory information into the cell state. In this work, another HBLSTM is implemented as the updating function to extract more information, and the NHBN is constructed via this nested hierarchical mechanism. In NHBN, two separate HBLSTMs (e.g., outer HBLSTM and inner HBLSTM) are applied to acquire the hidden state of the proposed NHBN model.

The input and the hidden state of the inner HBLSTM become:

$$\hat{\mathbf{h}}_{t-1} = \omega_{t1} \odot (\mathbf{f}_t \odot \mathbf{c}_{t-1}) \quad (16)$$

$$\hat{\mathbf{x}}_t = \omega_{t1} \odot (\mathbf{i}_t \odot \mathbf{g}_t) \quad (17)$$

Next, the hidden state of the inner HBLSTM is governed by the following:

$$\begin{aligned} \hat{\mathbf{f}}_t &= R(\mathbf{W}_f \hat{\mathbf{x}}_t + \mathbf{U}_f \hat{\mathbf{h}}_{t-1} + \mathbf{b}_f, \lambda) \\ \hat{\mathbf{i}}_t &= R(\mathbf{W}_i \hat{\mathbf{x}}_t + \mathbf{U}_i \hat{\mathbf{h}}_{t-1} + \mathbf{b}_i, \lambda) \\ \hat{\mathbf{o}}_t &= \sigma(\mathbf{W}_o \hat{\mathbf{x}}_t + \mathbf{U}_o \hat{\mathbf{h}}_{t-1} + \mathbf{b}_o) \\ \hat{\mathbf{g}}_t &= \varphi(\mathbf{W}_g \hat{\mathbf{x}}_t + \mathbf{U}_g \hat{\mathbf{h}}_{t-1} + \mathbf{b}_g) \\ \hat{\tilde{\mathbf{f}}}_t &= \text{cumsum}(\text{softmax}(\mathbf{W}_f \hat{\mathbf{x}}_t + \mathbf{U}_f \hat{\mathbf{h}}_{t-1} + \mathbf{b}_f)) \\ \hat{\tilde{\mathbf{i}}}_t &= 1 - \text{cumsum}(\text{softmax}(\mathbf{W}_i \hat{\mathbf{x}}_t + \mathbf{U}_i \hat{\mathbf{h}}_{t-1} + \mathbf{b}_i)) \\ \hat{\omega}_{t1} &= \hat{\tilde{\mathbf{f}}}_t \odot \hat{\tilde{\mathbf{i}}}_t \\ \hat{\omega}_{t2} &= \hat{\tilde{\mathbf{f}}}_t - \hat{\omega}_{t1} \\ \hat{\omega}_{t3} &= \hat{\tilde{\mathbf{i}}}_t - \hat{\omega}_{t1} \\ \hat{\mathbf{c}}_t &= \hat{\omega}_{t2} \odot \hat{\mathbf{c}}_{t-1} + \hat{\omega}_{t1} \odot (\hat{\mathbf{f}}_t \odot \hat{\mathbf{c}}_{t-1} + \hat{\mathbf{i}}_t \odot \hat{\mathbf{g}}_t) + \hat{\omega}_{t3} \odot \hat{\mathbf{g}}_t \\ \hat{\mathbf{h}}_t &= \hat{\mathbf{o}}_t \odot \varphi(\hat{\mathbf{c}}_t) \end{aligned} \quad (18)$$

The cell state of the outer HBLSTM is updated via the hidden state of the inner HBLSTM, and Eq. (15) is rewritten as follows:

$$\mathbf{c}_t = \omega_{t2} \odot \mathbf{c}_{t-1} + \hat{\mathbf{h}}_t + \omega_{t3} \odot \mathbf{g}_t \quad (19)$$

Based on the cell state of the outer HBLSTM and Eq. (3), the hidden state of the NHBN is computed as follows:

$$\mathbf{h}_t = \mathbf{o}_t \odot \varphi(\mathbf{c}_t) \quad (20)$$

Finally, the output of the NHBN model is calculated by:

$$\begin{aligned} \mathbf{y}_t^1 &= \text{Dropout}(\mathbf{h}_t) \\ \mathbf{y}_t^2 &= \text{Dropout}(\text{Relu}(\mathbf{W}_1 \mathbf{y}_t^1 + \mathbf{b}_1)) \\ \mathbf{y}_t^3 &= \mathbf{W}_2 \mathbf{y}_t^2 + \mathbf{b}_2 \\ \mathbf{y}_t &= \mathbf{W}_3 (\text{Dropout}(\mathbf{y}_t^3 + \mathbf{h}_t)) + \mathbf{b}_3 \end{aligned} \quad (21)$$

where $\mathbf{W}$ and $\mathbf{b}$ are weight and bias parameters, respectively, and $\mathbf{y}_t$ denotes the output of the NHBN model. Moreover, the specific model architecture of NHBN is listed in Table 2. "Input_num" denotes the number of neurons in input layer. "Hidden_num" denotes the number of neurons in NHBN cell. "Output_num" denotes the number of neurons in output layer. Obviously, the NHBN model has 9 layers, including one input layer, one NHBN cell layer, three full connection layers, three dropout layers, and one addition layer.

3.3. Proposed transfer gear RUL prediction method

There are two primary stages in the presented transfer RUL prediction method. In the first stage, the QFMDCAET is employed to construct gear's HIs from monitoring signals under various working conditions. In the second stage, the proposed NHBN model is built based on the transfer constructed gear HIs to estimate the RUL with the predicted gear HIs. The predicted HIs could represent the health status of the gear. The flowchart of the proposed transfer RUL prediction method is shown in Fig. 7. The pseudo code of this method is listed in Table 3 to better comprehend the developed transfer RUL prediction methodology.

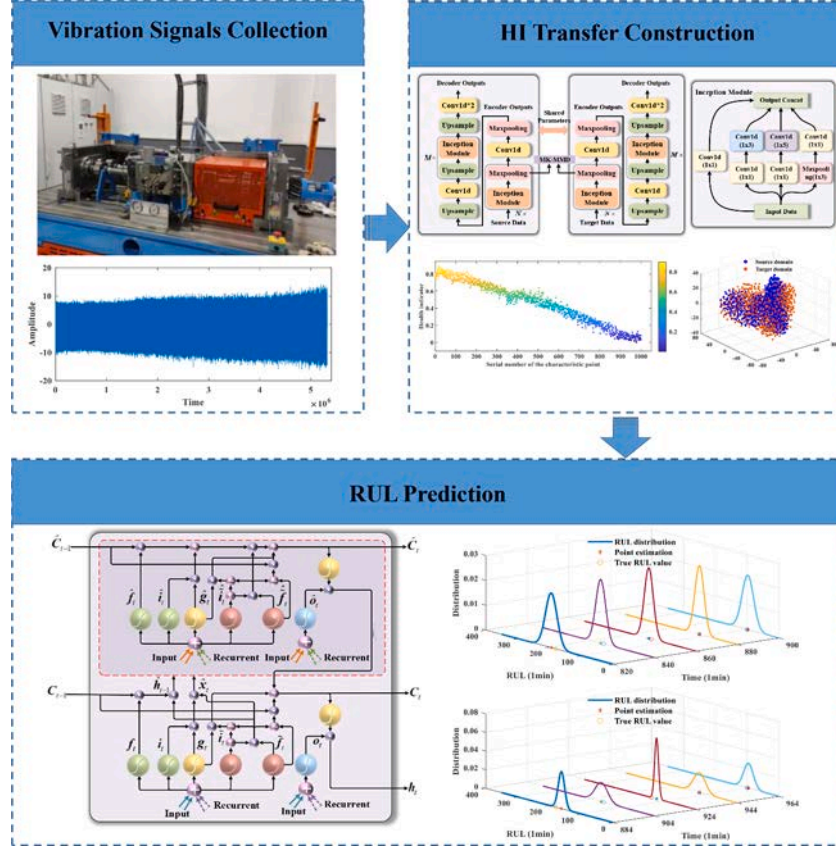
The specific procedures of the transfer gear RUL prediction method are as follows:

- Vibration signal collection:** The monitoring data containing the gear degradation information are acquired via some accelerometers under multiple operating conditions. The sampling frequency, sampling time, and sampling interval of the life-cycle samples are set as f_s , S_p , and S_q , respectively. The number of the full life-cycle data is N .
- HI Transfer construction:** The QFMDCAE network is constructed by the QFMDCAE network and inception module. Then, the pre-processed vibration signals of the source and target domains are fed into the constructed QFMDCAE to extract degradation features. The MK-MMD is utilized to minimize the distribution distance of the extracted source and target degradation features. The MK-MMD is defined as follows:

Table 2

The specific model architecture of the NHBN.

Module	Layer	Block	Operator	Neurons (Parameters)	Input size	Output size
NHBN Model	9	FC3	Fully-Connected	Output_num	Hidden_num	Output_num
	8	D3	Dropout	Dropout_rate	Hidden_num	Hidden_num
	7	A	Addition	Hidden_num	Hidden_num	Hidden_num
	6	FC2	Fully-Connected	Hidden_num	2*Hidden_num	Hidden_num
	5	D2	Dropout	Dropout_rate	2*Hidden_num	2*Hidden_num
	4	FC1	Fully-Connected	2*Hidden_num	Hidden_num	2*Hidden_num
	3	D1	Dropout	Dropout_rate	Hidden_num	Hidden_num
	2	NHBN	NHBN Cell	Hidden_num	Input_num	Hidden_num
	1	Input	1-D HI	Input_num	/	Input_num

**Fig. 7.** The flowchart of the proposed transfer RUL prediction approach.

$$L_{MK-MMD}(F_s, F_t) = \| E[\phi(F_s)] - E[\phi(F_t)] \|_{\mathcal{H}_k}^2, \quad (22)$$

where E is the mathematical expectation, $\phi(\cdot)$ is the continuous mapping relevant to the kernel function, F_s and F_t are the obtained abstract features from the source and target domains, respectively, and $\mathcal{H}_k$ is the RKHS with the specific kernel k . The kernel k is the convex combination of m different positive semidefinite kernels, which is defined as follows:

$$\mathcal{K} = \left\{ k = \sum_{i=1}^m \beta_i k_i : \sum_{i=1}^m \beta_i = 1, \beta_i \geq 0 \right\}. \quad (23)$$

Then, the monitoring data from the target domain are regarded as the input data of the trained QFMDCAE network to generate the gear's HI. A filtering method is employed to acquire the processed HI vector $\mathbf{H} = [H_1, H_2, H_3, \dots, H_L]^T$ and eliminate the outliers in the generated gear HI. The first l ($0 < l < L$) HI points $\mathbf{W} = [H_1, H_2, H_3, \dots, H_l]^T$ are applied to construct the training data for updating the parameters in NHBN.

- (1) **RUL prediction:** Based on the vector $\mathbf{W}$, a Hankel matrix is obtained as follows:

$$\mathbf{G} = \begin{bmatrix} H_1 & H_2 & \dots & H_{l-s} \\ H_2 & H_3 & \dots & H_{l-s+1} \\ \vdots & \vdots & \ddots & \vdots \\ H_{s+1} & H_{s+2} & \dots & H_l \end{bmatrix} = \begin{bmatrix} \mathbf{G}_1 \\ \mathbf{G}_2 \\ \vdots \\ \mathbf{G}_{s+1} \end{bmatrix} \quad (24)$$

where s denotes the number of neurons in the input layer, and:

$$\mathbf{G}_u = [H_u H_{u+1} \dots H_{l-s+u-1}] u = 1, 2, \dots, s+1. \quad (25)$$

The matrix $\hat{\mathbf{G}} = [\mathbf{G}_1 \mathbf{G}_2 \dots \mathbf{G}_s]^T$ is regarded as the input data of the NHBN, and the corresponding target data are the vector $\mathbf{G}_{s+1}$. The NHBN can be viewed as a learned nonlinear mapping function $\mathcal{F}$, and the loss function of the developed RUL prediction network is defined as follows:

Table 3

The pseudo code of the proposed transfer RUL prediction methodology.

Algorithm: Proposed transfer gear RUL prediction method	
Vibration signal collection:	
decimal%1	Collect monitoring data under multiple working conditions.
2	Sampling frequency is f_s , sampling time is S_p , sampling interval is S_q , the number of the full life-cycle data is N .
HI transfer construction:	
2	Pre-process vibration signals of the source and target domains.
2	Construct QFMDCAE network, and train the QFMDCAE via MK-MMD.
3	Obtain gear HI of target domain by the trained QFMDCAE network.
RUL prediction:	
2	Set the number of neurons in the input layer of NHBN is s .
2	The first l HI points in $\mathbf{W}$ are applied to form training data.
2	Construct Hankel matrix $\mathbf{G}$:
$\mathbf{G} = \begin{bmatrix} H_1 & H_2 & \cdots & H_{l-s} \\ H_2 & H_3 & \cdots & H_{l-s+1} \\ \vdots & \vdots & \ddots & \vdots \\ H_{s+1} & H_{s+2} & \cdots & H_l \end{bmatrix} = \begin{bmatrix} \mathbf{G}_1 \\ \mathbf{G}_2 \\ \vdots \\ \mathbf{G}_{s+1} \end{bmatrix}, \mathbf{G}_u = [H_u H_{u+1} \cdots H_{l-s+u-1}] u = 1, 2, \dots, s + 1$	
1	
2	The matrix $\hat{\mathbf{G}} = [\mathbf{G}_1 \mathbf{G}_2 \cdots \mathbf{G}_s]^T$ is utilized as input data of NHBN, the last vector $\mathbf{G}_{s+1}$ is used as output data of NHBN.
2	For number of epochs, do
2	Train NHBN model: the loss function is $\mathcal{L}(\mathbf{G}_{s+1}, \mathcal{T}(\hat{\mathbf{G}}))$.
3	Optimization method: real-time recurrent learning (RTRL) method.
2	End
2	If $\tilde{H}_{l-s+k-1} (\tilde{H}_{l-s+k-1} \in \tilde{\mathbf{G}}_k)$ does not reach the preset failure threshold, do
2	Consecutively calculate output vectors, where the k -th output vector $\tilde{\mathbf{G}}_k$ is defined as follows:
$\tilde{\mathbf{G}}_k = \mathcal{T}(\mathbf{G}_{k-s}, \mathbf{G}_{k-s+1}, \dots, \mathbf{G}_{s+1}, \tilde{\mathbf{G}}_{s+2}, \dots, \tilde{\mathbf{G}}_{s+(k-s-2)}, \tilde{\mathbf{G}}_{s+(k-s-1)})$	
2	Else if $\tilde{H}_{l-s+k-1} (\tilde{H}_{l-s+k-1} \in \tilde{\mathbf{G}}_k)$ reach the preset failure threshold, do
3	Calculate gear RUL via $(l - s + k - 1) \times S_q$.
4	End if
2	Output: the gear RUL.

$$\mathcal{L}(\mathbf{G}_{s+1}, \mathcal{T}(\hat{\mathbf{G}})) = \frac{1}{l-s} \sum_{i=1}^{l-s} [H_{s+i} - \mathcal{T}(H_i, H_{i+1}, \dots, H_{i+s-1})]^2 \quad (26)$$

To update the weight parameters in the proposed NHBN model, the real-time recurrent learning (RTRL) method is applied in this work. Based on the trained NHBN model, $\tilde{\mathbf{G}}_{s+2}$ the vector at the first prediction time (FPT) is calculated as follows:

$$\tilde{\mathbf{G}}_{s+2} = \mathcal{T}(\mathbf{G}_2, \mathbf{G}_3, \dots, \mathbf{G}_{s+1}) \quad (27)$$

Next, the matrix $\mathbf{G}$ can be updated as follows:

$$\mathbf{G} = \begin{bmatrix} H_1 & H_2 & \cdots & H_{l-s} \\ H_2 & H_3 & \cdots & H_{l-s+1} \\ \vdots & \vdots & \ddots & \vdots \\ H_{s+1} & H_{s+2} & \cdots & H_l \\ \tilde{H}_{s+2} & \tilde{H}_{s+3} & \cdots & \tilde{H}_{l+1} \end{bmatrix} = \begin{bmatrix} \mathbf{G}_1 \\ \mathbf{G}_2 \\ \vdots \\ \mathbf{G}_{s+1} \\ \tilde{\mathbf{G}}_{s+2} \end{bmatrix} \quad (28)$$

The output vector can be calculated step by step by repeating the above-mentioned steps. The k -th output vector is defined as follows:

$$\tilde{\mathbf{G}}_k = \mathcal{T}(\mathbf{G}_{k-s}, \mathbf{G}_{k-s+1}, \dots, \mathbf{G}_{s+1}, \tilde{\mathbf{G}}_{s+2}, \dots, \tilde{\mathbf{G}}_{s+(k-s-2)}, \tilde{\mathbf{G}}_{s+(k-s-1)}) \quad (29)$$

When the estimated value $\tilde{H}_{l-s+k-1} (\tilde{H}_{l-s+k-1} \in \tilde{\mathbf{G}}_k)$ reaches the preset failure threshold, the RUL of the gear is computed via $(l - s + k - 1) \times S_q$. The prediction performance of the proposed NHBN is verified by comparing the actual gear HI values.

4. Experimental results and analysis

4.1. Experimental data description

The whole life-cycle data of gear used in this work are collected on a

gear contact fatigue testing platform, which is made up of three main parts: the electrical control system part, the gear platform part, and the measurement part, as shown in Fig. 8. This gear test is an accelerated degradation test that contains two gearboxes (gearbox A and gearbox B), and two pairs of gears can be simultaneously tested under the same working condition.

The experimental datasets are described in Table 4. Furthermore, two speeds of 1100 rpm and 800 rpm are set to provide two experimental domains for RUL prediction under variable operating conditions. The experimental data are sampled at a frequency of 25 kHz, with a sampling time of 0.2 s per sample and a sampling interval of 60 s.

4.2. Transfer construction of gear's health indicator

To construct gears' HIs under variable operating conditions via the QFMDCAET framework, eight transfer tasks are constructed in this work. Table 5 demonstrates the specific settings of these transfer tasks under multiple working conditions. The source domain is the data for gear with labels in a single operation condition, while the target domain is the data for gear without labels in another operating condition. The HI construction method is trained using labeled data in the source domain and unlabeled data in the target domain. Specifically, task 3 is taken as a case to introduce the information of the hyper-parameters setting and training phase. The optimizer, batch size, epoch, and learning rate are Adam, 128, 150, and 0.005, respectively. In the training phase, the source domain data and target domain data are fed into the QFMDCAET framework at the same time for domain invariant feature extraction. In addition, the weight parameters of the network are shared for the source domain data and the target domain data.

The constructed gear HIs for the eight tasks are shown in Fig. 9. The obtained HIs in the target domain have superior monotonicity, and HIs can vividly reflect the gear degradation trends over time. For example, in the initial stage of the experiment, the degradation rate is low because there are no defects on the tooth surface of the tested gear. In the middle stage of the experiment, the degradation rate becomes larger than that in the early stage. This is because some defects appear on the tooth surface of the tested gear. In the final stage of the experiment, serious defects lead to the tested gear being in the stopped state, and the degradation rate is at its highest. Considering the constructed HIs for tasks 3, 5, 6, 7, and 8, the failure threshold for RUL prediction can be uniformly preset. Finally, the constructed HIs show that the QFMDCAET framework is capable of generating HIs under different working conditions.

To further show the superiority of the proposed QFMDCAET framework, the QFMDCAE with MMD (QFMDCAE-MMD) and the QFMDCAE without transfer learning techniques are used for comparison. The HI evaluation indexes, including monotonicity (Mon), correlation (Cor), and robustness (Rob), are employed to assess the construction HIs. Moreover, a composite index (CI) [16] that combines Mon, Cor, and Rob is also utilized to comprehensively evaluate the gear HIs. The comparative results are demonstrated in Fig. 10. It can be seen that the QFMDCAET is more suitable for gear HI construction than the QFMDCAE-MMD and QFMDCAE methods. Finally, it can be concluded that the obtained HIs based on the presented HI construction framework are more appropriate for RUL prediction.

To further validate the effectiveness of the QFMDCAET framework for HI construction under variable working conditions, the extracted features of the penultimate max-pooling layer in the encoder of QFMDCAE are visualized and analyzed in this work. The t-SNE algorithm is applied to visualize the high-dimensional source and target features reduced to three-dimensional space by taking task 3 as an example, and the results are demonstrated in Fig. 11. It can be observed that features extracted by QFMDCAET exhibit a flow structure in space. This phenomenon is because the input data are the full life-cycle data of the gear, implying the change process of the gear from a healthy state to a failure state. This further illustrates the capacity of the QFMDCAET to reflect the complicated mapping relationship between the monitoring

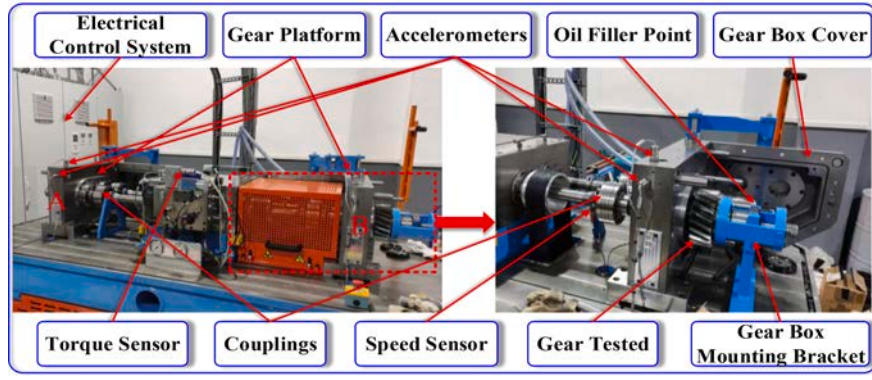


Fig. 8. Gear contact fatigue testing platform.

Table 4

The description of the experimental datasets.

Operating condition	Condition1	Condition2
Rotating speed (rpm)	1100	800
Torque (Nm)	4400	4070
Gearbox A	GearA1-1	GearA2-1
	GearA1-2	GearA2-2
Gearbox B	GearB1-1	GearB2-1
	GearB1-2	GearB2-2

Table 5

Cross-domain HI construction tasks for gear datasets.

Tasks	Source domain(Labeled)	Target domain(Unlabeled)
Task 1	GearA1-1	GearA2-1
Task 2	GearA1-2	GearA2-2
Task 3	GearB1-1	GearB2-1
Task 4	GearB1-2	GearB2-2
Task 5	GearA2-1	GearA1-1
Task 6	GearA2-2	GearA1-2
Task 7	GearB2-1	GearB1-1
Task 8	GearB2-2	GearB1-2

data and the HI. Furthermore, as can be seen from Fig. 11, the source and target features are largely aligned in space. The feature distributions of the source domain and the target domain are approximately aligned, and the source and target features intermingle with each other. Furthermore, the feature trends of the source domain and target domain in Fig. 11 are approximately the same. Therefore, the proposed QFMDCAET to achieve feature alignment in the source and target domains is reasonable and effective.

4.3. RUL prediction and uncertainty quantification

The transfer-constructed HIs for task 3 and task 7, denoted as THI-3 and THI-7, are utilized to verify the prediction performance of the

proposed RUL prediction approach. The lengths of THI-3 and THI-7 are 1000 and 1064, respectively. The proposed NHBN model is applied to predict RULs at different prediction points, including 100, 120, 140, 160, and 180 prediction points. The first 900 HI points are utilized to construct the training data for training the NHBN model by taking 100 prediction points and THI-3 as an example. Then, the last 100 HI points are estimated via the trained NHBN. The predicted HI points can describe the health state of the gear over time. The actual RUL is 100 min, and the uniform failure threshold is set as 0, meaning that the gear has completely failed. The hyper-parameters are selected via grid search method. The number of neurons in input layer, the number of neurons in NHBN cell, learning rate, dropout rate, and the number of neurons in output layer are 120, 21, 0.001, 0.01, and 1, respectively. In the training phase, stochastic gradient descent algorithm is applied to update the weight parameters in NHBN, and the training epoch is 200. The proposed transfer RUL prediction method is implemented using Pytorch platform with i7-10700F processor, NVIDIA GeForce RTX 3080 graphics card, and 16 G memory. The number of parameters of QFMDCAE and NHBN, the average model training time, and the average model test time are calculated to analyze the computational complexity of the proposed life transfer prediction approach. The RULs of gearB2-1 and gearB1-1 at different prediction points are obtained via the trained NHBN with the optimized hyper-parameters. The trained NHBN is implemented 50 times on THI-3 and THI-7 to guarantee that the predicted RULs are reliable and fair. The mean value of these estimated RULs is used as the final RUL. Moreover, the uncertainty of NHBN is quantified by opening the dropout layer during testing. The interval estimations, actual RULs, and predicted RULs of gearB2-1 and gearB1-1 at different prediction points are shown in Table 6. The probability density functions (PDFs) of gearB2-1 and gearB1-1 at different prediction points are illustrated in Fig. 12.

According to Table 6 and Fig. 12, the predicted RULs are close to the actual RULs at different prediction points, and the PDFs are concentrated on the predicted RULs without large uncertainty. Therefore, the maintenance decision based on uncertainty can be performed to avoid some disaster accidents. In addition, the uncertainty denoted by the PDF

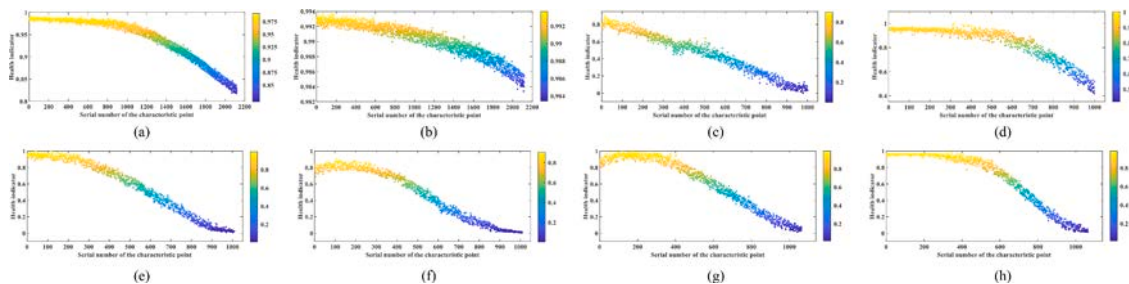


Fig. 9. Constructed gear HIs: (a) task 1, (b) task 2, (c) task 3, (d) task 4, (e) task 5, (f) task 6, (g) task 7, (h) task 8.

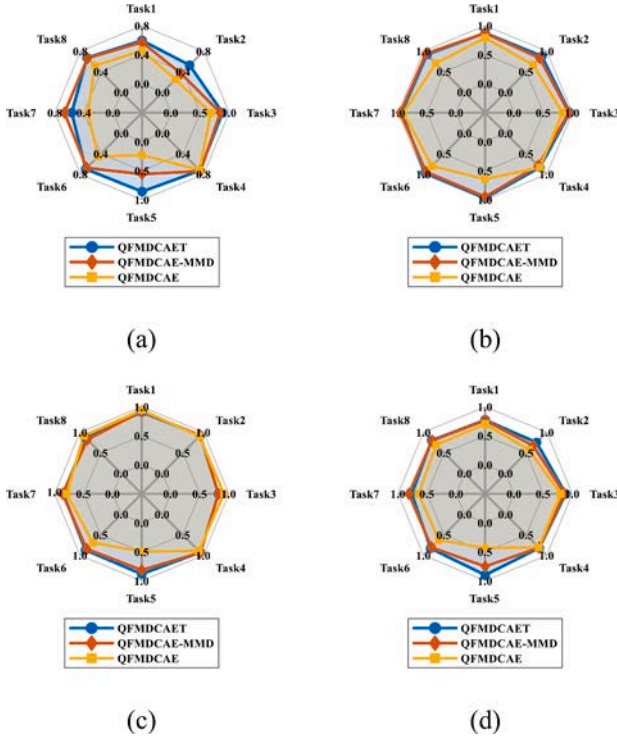


Fig. 10. The evaluation indexes of different gear HIs for different tasks: (a) Mon, (b) Cor, (c) Rob, (d) CI.

of RULs can be interpreted as how much confidence the proposed approach has in the RUL prediction. Moreover, the PDF of RULs with small dispersion indicates that the proposed RUL prediction method is confident regarding RUL estimation. It can be concluded that the Gumbel function and the nested hierarchical mechanism are effective in improving the RUL prediction accuracy. QFMDCAE and NHBN, respectively have 25,158 and 142,869 trainable parameters. The average training time of QFMDCAE and NHBN are 813.40 and 238.08 s, respectively. The average test time of QFMDCAE and NHBN are 1.25 and 0.07 s, respectively. The short test time of the proposed method

demonstrates that the trained model can be applied to the online gear life prediction, so as to guarantee the safety and reliability of equipment.

4.4. Comparison with other RUL prediction methods

To illustrate the superior prediction performance of the proposed NHBN model, six RUL prediction methods, including LSTM, gated recurrent unit (GRU), macro-micro attention mechanism LSTM (MMALSTM) [12], ON-LSTM [27], gated attention unit (GAU) [4], and gated dual attention unit (GDAU) [4] are employed for comparison with the proposed NHBN model. The hyper-parameters of these methods are the same as those of the proposed NHBN to ensure the fairness and rationality of the comparison results. In addition, the mean absolute error (MAE), normalized root mean square error (NRMSE), mean absolute percentage error (MAPE), root mean square error (RMSE), and score are applied to evaluate the prediction performance of these RUL prediction methods. The comparative results of the above five evaluation indexes obtained using the predicted RULs of gearB2-1 and gearB1-1 at different prediction points are shown in Table 7.

According to Table 7, the proposed RUL prediction method has the best prediction performance among these methods. Compared with other methods, it has lower prediction error and better stability. The prediction accuracy can still reach 89.89% when the number of prediction points is increased to 180, indicating that the proposed NHBN method is capable of achieving long-term RUL prediction. The above results demonstrate that the Gumbel function and the nested hierarchical mechanism can improve the memory capacity of NHBN for RUL prediction at different prediction points, especially for long-term memory ability.

Moreover, according to the evaluation results, the prediction performance of ON-LSTM is better than that of LSTM, GRU, MMALSTM, GAU, and GDAU. This phenomenon indicates that the hierarchical mechanism in ON-LSTM can leverage the sequence information from the current input data and historical data to update memory information in the cell state of ON-LSTM. According to the evaluation results of MMALSTM, GAU, and GDAU, the attention mechanism can improve the accuracy of prediction to some extent, but the prediction stability is not as good as that of ON-LSTM and NHBN. When the prediction point increases, the prediction performance of these attention mechanism-based methods may dramatically decrease since the attention mechanism

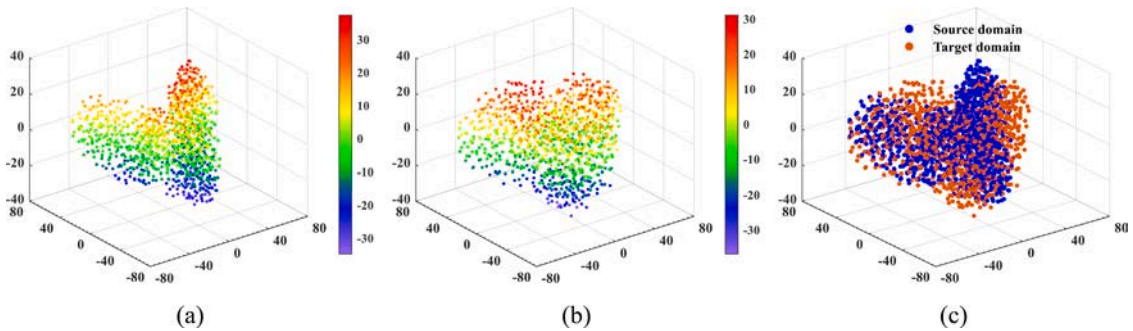


Fig. 11. Degradation feature alignment results for source and target domains: (a) source domain, (b) target domain, (c) source and target domains.

Table 6

RUL prediction with uncertainty at different prediction points for gears.

GearB2-1				GearB1-1			
Prediction point	Actual RUL	Predicted RUL	95% confidence interval	Prediction point	Actual RUL	Predicted RUL	95% confidence interval
100	100	98.06	[62.32 133.80]	100	100	96.10	[66.74 125.46]
120	120	119.64	[91.35 147.93]	120	120	119.80	[83.28 156.32]
140	140	148.44	[120.91 175.97]	140	140	124.12	[112.14 136.10]
160	160	169.30	[138.80 199.80]	160	160	167.30	[129.37 205.23]
180	180	194.38	[158.76 230.00]	180	180	162.40	[142.72 182.08]

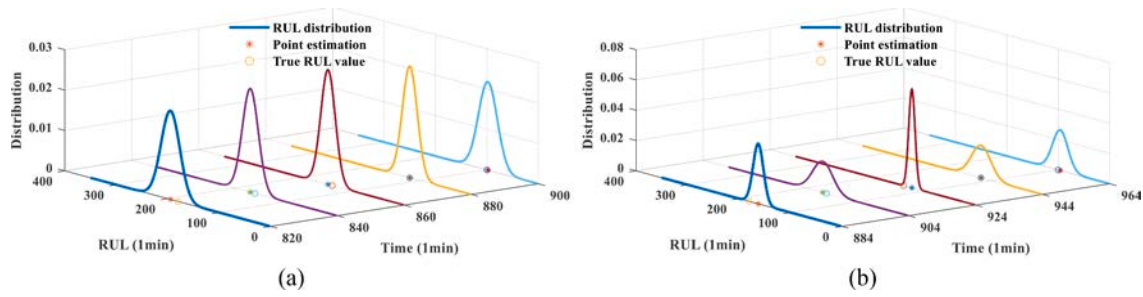


Fig. 12. RUL prediction with uncertainty quantification for gears: (a) gearB2-1, (b) gearB1-1.

Table 7

The evaluation results of different methods at different prediction points.

Prediction point	Method	MAE	NRMSE	MAPE	RMSE	Score
100	LSTM	33.4100	0.3753	33.41%	49.2781	0.2886
	GRU	35.9700	0.3902	35.97%	39.8897	0.2009
	MMALSTM	42.9100	0.4359	42.91%	61.3409	0.2835
	ON-LSTM	16.7300	0.2316	16.73%	19.6568	0.5381
	GAU	71.8600	2.7357	71.86%	76.9831	0.1253
	GDAU	40.1500	0.7572	40.15%	45.3466	0.3172
	NHBN	12.7000	0.1731	12.70%	16.8036	0.5059
	LSTM	76.1000	2.0041	63.42%	87.9782	0.2164
120	GRU	37.6500	0.4736	31.38%	39.0047	0.3510
	MMALSTM	76.6900	2.0366	63.91%	88.2040	0.2099
	ON-LSTM	22.1200	0.2662	18.43%	26.3431	0.5392
	GAU	54.3000	0.8446	45.25%	55.4931	0.2193
	GDAU	51.7500	0.7689	43.13%	52.4781	0.2312
	NHBN	12.8000	0.1378	10.67%	16.5033	0.4989
	LSTM	27.5100	0.2896	19.65%	33.9445	0.5135
	GRU	37.0600	0.3739	26.47%	38.4910	0.4112
140	MMALSTM	28.1400	0.2817	20.10%	34.1573	0.4614
	ON-LSTM	27.8200	0.2638	19.87%	29.6014	0.5177
	GAU	52.5600	0.6166	37.54%	53.9188	0.2839
	GDAU	76.4400	1.2094	54.60%	76.8690	0.1539
	NHBN	14.9000	0.1221	10.64%	16.6343	0.5467
	LSTM	98.8600	1.1976	61.79%	116.8855	0.0700
	GRU	31.2600	0.2548	19.54%	44.0236	0.5137
	MMALSTM	99.7200	1.1797	62.33%	117.0071	0.0648
160	ON-LSTM	23.0000	0.1493	14.38%	26.6908	0.2772
	GAU	33.7700	0.3239	21.11%	42.0213	0.4973
	GDAU	48.5100	0.4416	30.32%	49.2369	0.3552
	NHBN	16.1400	0.1146	10.09%	19.2883	0.4540
	LSTM	47.4500	0.3784	26.36%	50.1598	0.4208
	GRU	67.5100	0.6072	37.51%	68.3064	0.2779
	MMALSTM	46.2100	0.3640	25.67%	48.7018	0.4292
	ON-LSTM	35.2100	0.2537	19.56%	36.7384	0.5183
180	GAU	90.8700	1.0277	50.48%	91.5949	0.1780
	GDAU	91.9800	1.0476	51.10%	92.2088	0.1715
	NHBN	18.1900	0.1215	10.11%	21.6686	0.5707

cannot learn the sequence information in these methods. According to the above analysis, the developed RUL prediction method has superior prediction performance and stability due to the Gumbel function and nested hierarchical mechanism in NHBN.

To additionally verify the effectiveness of the Gumbel function in the

NHBN method, the output values of the input and forget gates in the trained NHBN and LSTM are obtained for 180 prediction points in gearB1-1. The heatmap of these gating values is demonstrated in Fig. 13. The gating values of LSTM are relatively uniform, and the gating values of NHBN are concentrated in the boundary regions. This phenomenon

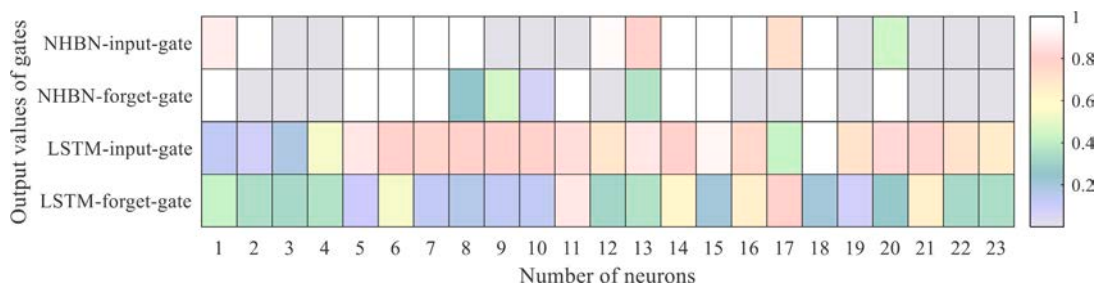


Fig. 13. The heatmap of the gating values of the trained NHBN and classic LSTM.

shows that the Gumbel function meets the expectation and successfully brings the gating values closer to 0 or 1.

In order to quantify the data distributions of the gating values in NHBN and traditional LSTM, the specific proportions of different value intervals of the input and forget gates are shown in Fig. 14, where it can be observed that most gating values of NHBN are at the boundary regions. For the gating values of input and forget gates of NHBN, 45% and 55% of the gating values are concentrated around 0, and 35% and 30% of the gating values are concentrated around 1. In contrast, considering the gating values of input and forget gates of LSTM, only 4% of the gating values are concentrated around 0, and 9% of the gating values are concentrated around 1. This observation indicates that the Gumbel function can generate additional gating values around 0 or 1, which is aligned with the design logic of the gate mechanism. The improved binary-valued gates in NHBN can enhance the long-term prediction performance of the proposed RUL prediction method.

4.5. Ablation experiment

To further validate the effectiveness of the Gumbel function and nested hierarchical mechanism in the proposed method, an ablation experiment is implemented. The nested hierarchical LSTM (NHLSTM) without the Gumbel function and HBLSTM without nested hierarchical mechanism are constructed to estimate the RUL of gearB1–1 at 180 prediction points. Moreover, the hyper-parameters of the above networks are set as same as those of the proposed NHBN method. The results of the evaluation indexes are presented in Table 8.

According to Table 8, the proposed NHBN has the highest score among these RUL prediction methods. Considering MAE, NRMSE, MAPE, and RMSE, the NHBN has the lowest prediction error, indicating that it has a superior prediction performance than NHLSTM and HBLSTM. Moreover, the MAE and MAPE of NHLSTM are lower than those of HBLSTM, and the score of NHLSTM is higher than that of HBLSTM. This is because the nested hierarchical mechanism is not applied in HBLSTM, so the sequence information is not effectively used in HBLSTM for long-term memory information learning. Finally, the ablation results demonstrate that the developed NHBN method that combines the Gumbel function and nested hierarchical mechanism is effective for RUL estimation, especially for long-term RUL prediction.

5. Conclusions

In this work, a novel transfer life prediction method was proposed to achieve remaining useful life (RUL) prediction under different operation conditions. The QFMDCAET framework was first used to obtain cross-condition HIs with significant degradation trends, and the failure threshold indispensable for RUL prediction was uniformly determined based on the constructed gear HIs.

The NHBN was developed for estimating gear RULs under multiple working conditions. In this NHBN model, a Gumbel function that can approximate the sampled values of the Bernoulli distribution was designed to obtain more gating values around 0 or 1, improving the long-term memory capacity of NHBN. Based on the Gumbel function and a hierarchical mechanism inspired by ON-LSTM, an outer HBLSTM network was presented for dividing the sequence information of cell state into high cell hierarchy, middle cell hierarchy, and low cell hierarchy. To fully utilize the sequence information in the middle cell hierarchy of the outer HBLSTM, a nested hierarchical mechanism that uses the hidden state of an inner HBLSTM to update the cell state in the middle cell hierarchy of the outer HBLSTM was built. The NHBN model, which has an exceptional memory capacity, was constructed for RUL prediction at different prediction points using the outer HBLSTM and the inner HBLSTM. The effectiveness of the proposed RUL prediction approach was verified by the life-cycle datasets of gears. Moreover, the PDFs of the predicted RULs were obtained to quantify the uncertainty of the proposed RUL prediction method. Finally, the comparative results

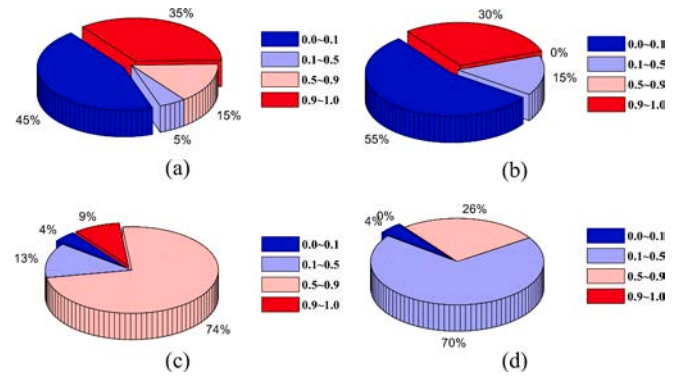


Fig. 14. The proportions of the different value intervals of the input and forget gates in NHBN and LSTM: (a) input gate of NHBN, (b) forget gate of NHBN, (c) input gate of LSTM, (d) forget gate of LSTM.

Table 8

The evaluation results for the ablation experiment.

Method	MAE	NRMSE	MAPE	RMSE	Score
NHLSTM	21.1600	0.1500	11.76%	23.8302	0.6793
HBLSTM	22.1200	0.1493	12.29%	23.5661	0.6612
NHBN	17.6000	0.1245	9.78%	20.2119	0.7250

demonstrated that the NHBN is more effective and advanced than other existing methods in RUL prediction, especially for long-term RUL prediction.

CRedit authorship contribution statement

Dingliang Chen: Methodology, Formal analysis, Writing – original draft. **Wei Cai:** Data curation, Visualization. **Hangjun Yu:** Investigation, Validation. **Fei Wu:** Data curation, Software. **Yi Qin:** Conceptualization, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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